Majid Haji Bagheri

R&D Engineer · Device Fabrication & Characterization · Energy Harvesting & Storage · MEMS / Micro-Nanofabrication · Hardware–ML Integration

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Professional Summary

R&D engineer and PhD candidate (ECE, University of Waterloo) with 6+ years designing, fabricating, and characterizing energy-harvesting and storage devices. Full-cycle cleanroom microfabrication, device modeling, simulation, and instrumentation. Built PENGs/TENGs and Li-ion anodes; integrated sensors with Python ML for signal classification. 3+ years leading industry-collaborative R&D (NRC, Shimco), translating partner objectives into delivered prototypes, reports, and presentations. First author of multiple peer-reviewed papers; co-author in Nature Communications. Expected PhD: Fall 2026.

Professional Experience

Graduate Research Assistant (PhD Candidate) · University of Waterloo — Prof. Dayan Ban's Lab, Waterloo, ON

(Sep 2022 – Present)

- Led three concurrent industry-collaborative R&D projects with the National Research Council of Canada (NRC) and Shimco North America, from planning through delivery — producing milestone reports, technical presentations, and peer-reviewed publications.
- Designed, fabricated, and characterized high-performance piezoelectric and triboelectric nanogenerators (PENGs/TENGs) for self-powered sensing, wearable electronics, and energy harvesting; results published in Advanced Functional Materials, Nano Energy, and Energy & Environmental Materials.
- Built a TENG-based acoustic sensor integrated with a Python machine-learning classifier for live sound monitoring and captioning; published in Advanced Sensor Research and presented at IEEE CCECE.
- Developed multi-layer cascaded PENG architectures enabling reliable operation under cryogenic and low-temperature conditions, expanding the device deployment envelope.
- Performed first-principles DFT simulations (Quantum ESPRESSO) on lead-free perovskites to investigate electronic structure and ferroelectric polarization, guiding material selection.
- Fabricated, simulated, and tested InGaN/GaN multiple-quantum-well PIN micro-photodetectors with on-chip grating couplers in the Quantum-Nano Fabrication & Characterization Facility (QNFCF) cleanroom, following a ~30-step multilayer process flow covering mesa definition, passivation, metallization, and top-cladding integration.
- Executed the full process-integration toolbox at QNFCF: PECVD and ALD dielectric deposition, e-beam evaporation and sputter metallization (Cr/Au, Al), hard-mask patterning, reactive-ion etching of metals, oxides, and ITO, lift-off, and wet processing (piranha cleaning, KOH treatment, wet metal etching) with photoresist and e-beam-resist workflows.
- Trained on and worked with electron-beam lithography, maskless laser lithography, plasma deposition and dry-etch cluster tools, resist spin coating, and solvent wet-bench processing at QNFCF.
- Investigated ferroelectric materials and their low-temperature phase transitions through experimental characterization and theoretical simulations using the Ferret MOOSE framework, focusing on phase-field modeling and domain evolution.
- Developed user interfaces (GUI) for lab-instrument control and automation, saving measurement time and resources.

Graduate Research Assistant (MSc) · Sharif University of Technology, Tehran, Iran (Feb 2021 – Feb 2022)

- Led the battery research group: experimental planning, project workflows, and student-team coordination for lithium-ion anode development.
- Raised Li-ion anode performance ~50% and roughly doubled cycle life with a binder-free SnS_2 anode using $Ti_3C_2T_x$ MXene as a conductive binder.
- Synthesized SnS_2 nanostructured anodes; fabricated freestanding MXene and graphene-oxide papers; optimized binder systems (PEDOT:PSS, PVDF, CMC).
- Designed, built, and programmed a custom in-house glove box for oxygen- and moisture-sensitive cell fabrication, including sealing, atmosphere control, and instrumentation.
- Delivered industry milestones well ahead of schedule (~8 vs 18 months; 12 vs 24 months) while exceeding performance targets (~1.8× battery life).

Undergraduate Research Assistant · Sharif University of Technology, Tehran, Iran (Nov 2019 – Feb 2021)

- Synthesized SnS_2–graphene-oxide composites for Li-ion anodes; produced graphene oxide via Hummers' method and $Ti_3C_2T_x$ MXene via LiF/HCl etching.
- Industry collaboration (Parthian Battery Novin & Iran Nano Battery Network): contributed to a Sharif–industry R&D project developing high-performance Li-ion batteries with self-healing nanomaterials ($Ti_3C_2T_x$ MXenes) to improve cycling stability and mechanical resilience.
- Investigated MXene-based composite anodes and binder formulations to suppress capacity fade, mitigate volumetric expansion, and improve Coulombic efficiency.
- Coordinated experimental planning between the academic lab and industry partners; performed materials/device characterization (XRD, FTIR, SEM/EDX, TEM, I–V, C–V, CV, battery cycling).

Undergraduate Researcher · Iran University of Science & Technology, Tehran, Iran (Feb 2018 – Aug 2018)

- Implemented stochastic Monte Carlo simulations of the ballistic-deposition model to study non-equilibrium thin-film growth, surface-morphology evolution, and dynamic roughening on 1+1D and 2+1D substrates.
- Computed roughness scaling exponents and verified Family–Vicsek dynamic scaling, confirming Kardar–Parisi–Zhang (KPZ) universality-class behavior.
- Built foundational PVD and thin-film-growth understanding later applied to sputtering, CVD, and cleanroom fabrication; programmed simulation and visualization workflows in Python.

Technical Skills

Device Fabrication: PENG/TENG and photodetector fabrication; III-nitride (InGaN/GaN MQW) device processing; Li-ion cell assembly; photolithography; electron-beam lithography; maskless laser lithography; thin-film deposition (sputtering, e-beam evaporation, PECVD, ALD, CVD, PVD, CVT); resist spin coating (photo & e-beam resists); reactive-ion etching (RIE); lift-off; wet processing (piranha, KOH, solvent wet bench); electrospinning; spray & vacuum-filtration coating; MEMS/CMOS/diode/MOSFET fabrication; bonding; class-controlled cleanroom operation (incl. QNFCF).

Characterization: XRD; FTIR; SEM/EDX; TEM; AFM; DSC; DMTA; TGA; optical microscopy; I–V; C–V; cyclic voltammetry; galvanostatic battery cycling; electromechanical & electroacoustic testing.

Simulation & Modeling: Quantum ESPRESSO (DFT, electronic structure, ferroelectric polarization); Ferret/MOOSE phase-field modeling of ferroelectric materials, domain evolution, and temperature-dependent phase transitions; COMSOL Multiphysics; Ansys Lumerical; Monte Carlo / stochastic thin-film growth (KPZ-class); MATLAB.

Programming & Data: Python (signal processing, ML classification); MATLAB; Origin Pro; LaTeX; Microsoft Office.

Custom Instrumentation: Motor-tapping energy-harvester test benches; cryogenic / low-temperature test rigs; oxygen-controlled glove boxes.

Leadership & Collaboration: Industry/government project planning; milestone reporting; technical presentations (NRC, Shimco); mentoring of co-op and undergraduate research assistants.

Certification: Certified Radiation Protection Professional — Non-Destructive Testing (NDT), Medical & Industrial.

Education

PhD, Electrical & Computer Engineering (Nanotechnology) — University of Waterloo, Waterloo, ON, Canada

(Sep 2022 – Fall 2026 (Expected))

MSc, Physics — Sharif University of Technology, Tehran, Iran (Sep 2018 – Feb 2021)

BSc, Physics (Condensed Matter) — Iran University of Science & Technology, Tehran, Iran (Sep 2014 – Sep 2018)

Selected Publications & Recognition

- First author, "Lone-Pair-Induced Lattice Distortion in Lead-Free Hybrid Perovskites for High-Performance Nanogenerators," Advanced Functional Materials (2026).
- Co-author, "Breaking Dielectric Dilemma via Polymer-Functionalized Perovskite Piezocomposite," Nature Communications, 15, 9511 (2024).
- First author, "Self-Healable, Recyclable, Mechanically Robust Vitrimer Composite for TENGs and Self-Powered Wireless Electronics," Nano Energy (2024).
- First author, "Sustainable Live Sound Monitoring & Classification via TENG and Machine Learning," Energy & Environmental Materials (2025).
- 17 peer-reviewed works; one manuscript under review; featured by CTV News Kitchener & UWaterloo News.